Wireless 03 | Smart System Upgrade (SSU)

Overview

Arista's Smart System Upgrade (SSU) is a feature to minimize traffic loss when upgrading from one SSU-supported EOS version to a newer SSU-supported EOS version. SSU is also referred to as 'hitless' upgrades. The SSU feature allows a switch to maintain packet forwarding (Data Plane) while the Management and Control plane perform the OS upgrade.

What you'll learn:

- Identify SSU prerequisites and configuration requirements
- Validate switch readiness for a hitless firmware upgrade
- Perform a Smart System Upgrade using the CLI
- Verify zero-downtime operation during the upgrade process

> **Arista Smart System Upgrade**
>
> Additional information about this feature can be found in the Arista TOI for Smart System Upgrade

In our workshop lab topology you will see that each leaf in your pod is directly connected to the access point and RaspberryPi client. Traditionally, a firmware upgrade on the lead in the pod would cause the access point, wireless clients connected to the access point, and the raspberry pi client to lose network connectivity. In this lab, we will use Arista SSU on the leaf switch in your pod to perform a firmware upgrade minimizing any network disruptions for both wired and wireless clients.

Prerequisites

- Continuous POE should be configured to maintain POE power delivery to connected devices.
- Must be running an EOS version that includes the SSU feature.
- Must be upgrading to a new EOS version that also includes the SSU feature.
- Spanning-tree must be running in MST mode or disabled.

- Spanning-tree edge ports must have portfast and BPDUGuard enabled.
- Switches running BGP must be configured with `graceful-restart` otherwise routes are lost and the hardware may fail to forward traffic.

Caveats

As you can imagine, disconnecting the Control and Management plane off the data plane will come with some caveats depending on what features you are running! While that's the case, most instances where SSU is valuable is where we have no resiliency or ability to "route around" the switch in maintenance.

- SSU only supports upgrades. Hitless image downgrades are not supported.
- If a new EOS version includes an FPGA upgrade, the FPGA upgrade will be suppressed. FPGA upgrades require a full reboot of a switch to apply.
- Some switch features, when in use, will prevent SSU from starting. See the Arista TOI for more details

Open Launchpad | **Login Help** | **Lab Topology**

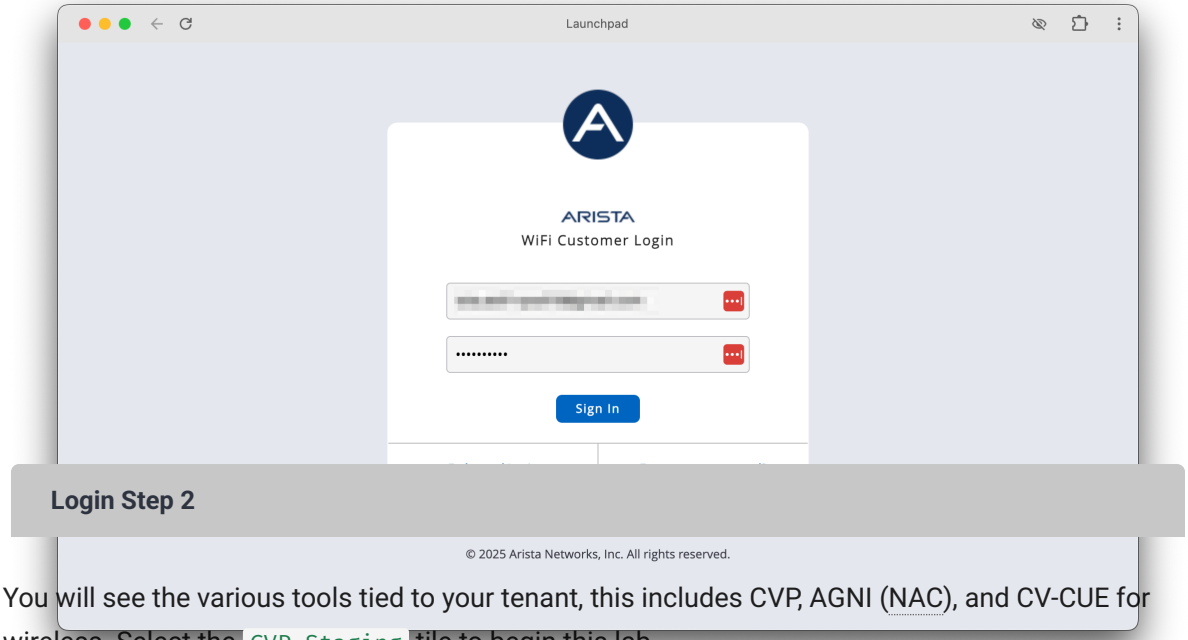
CloudVision Login

If you're not already logged into CloudVision (CVaaS), navigate to the Arista Launchpad for your lab.

Open Launchpad

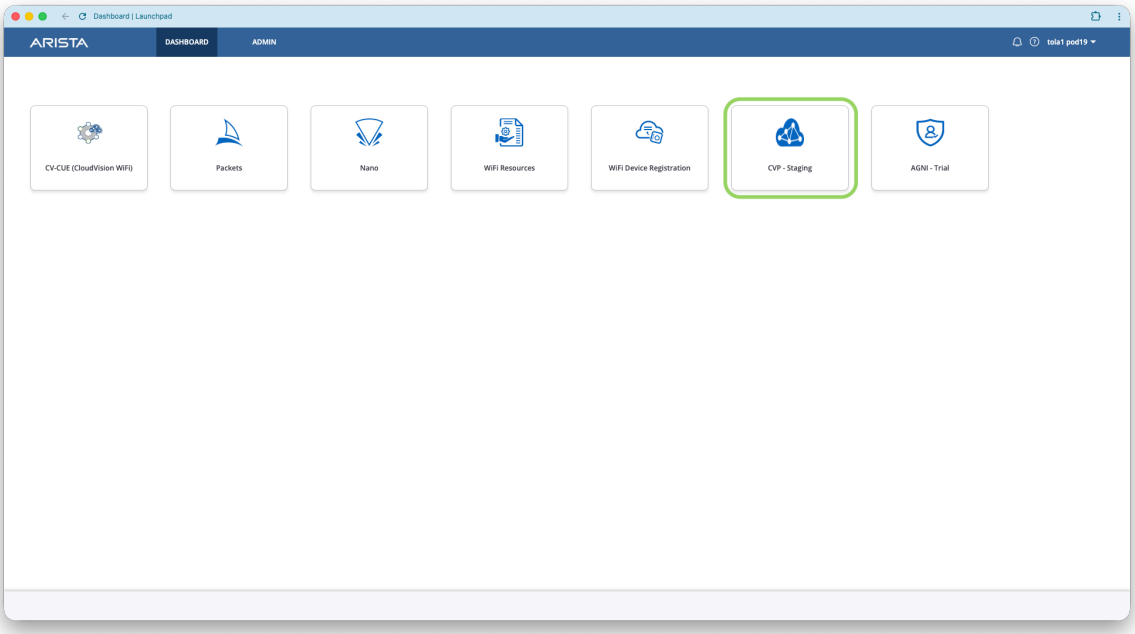
Login Step 1

Select the `Log in with Launchpad` button and provide your assigned lab#+pod## email address and password:



Login Step 2

You will see the various tools tied to your tenant, this includes CVP, AGNI (NAC), and CV-CUE for wireless. Select the `CVP-Staging` tile to begin this lab.



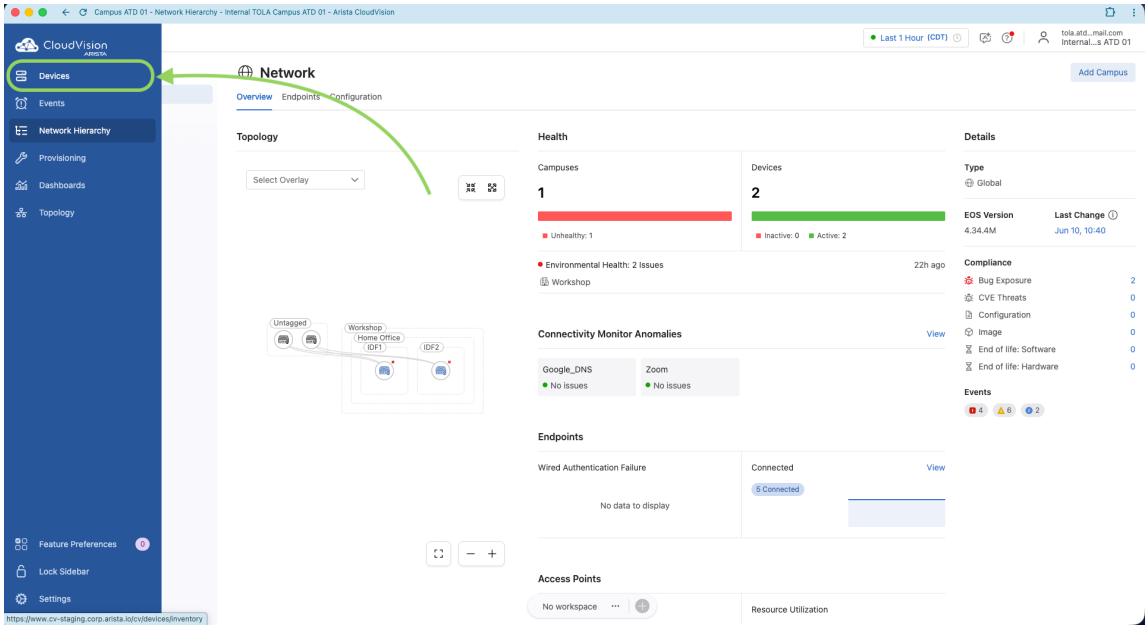
Prepping for SSU

Let's begin the hands-on portion of this lab. SSU can be triggered on the command line or via CloudVision. For this lab, we will perform a SSU upgrade on your switch while monitoring pings

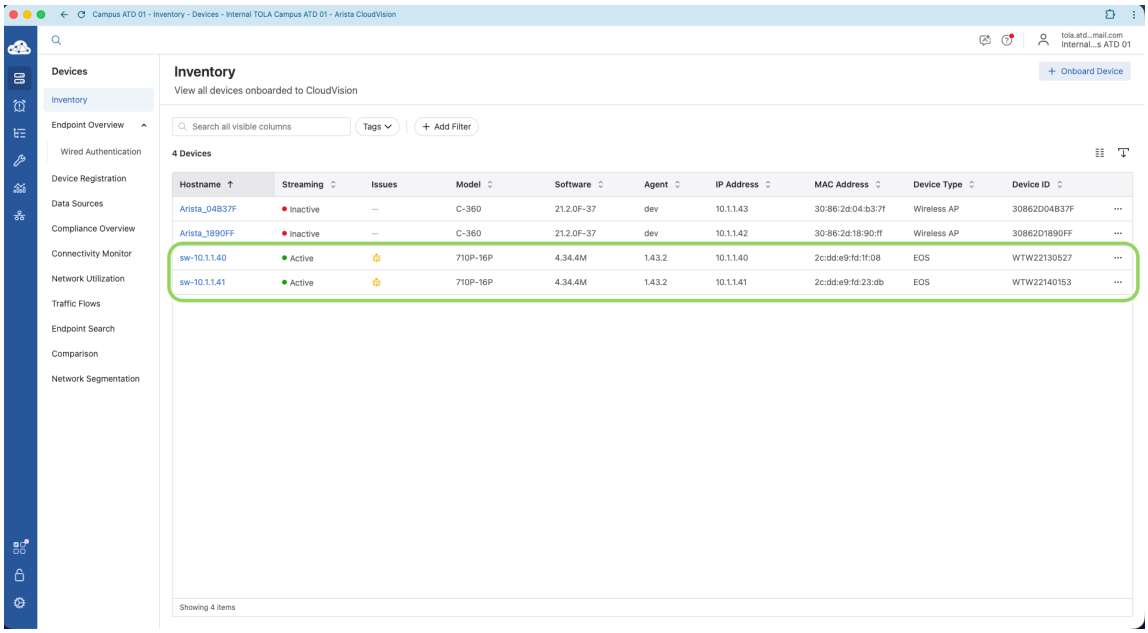
from the RaspberryPi.

> **Who needs a console cable??**
>
> Using the console we will get a more in depth look at the logs as the switch upgrades. However, it is not required.

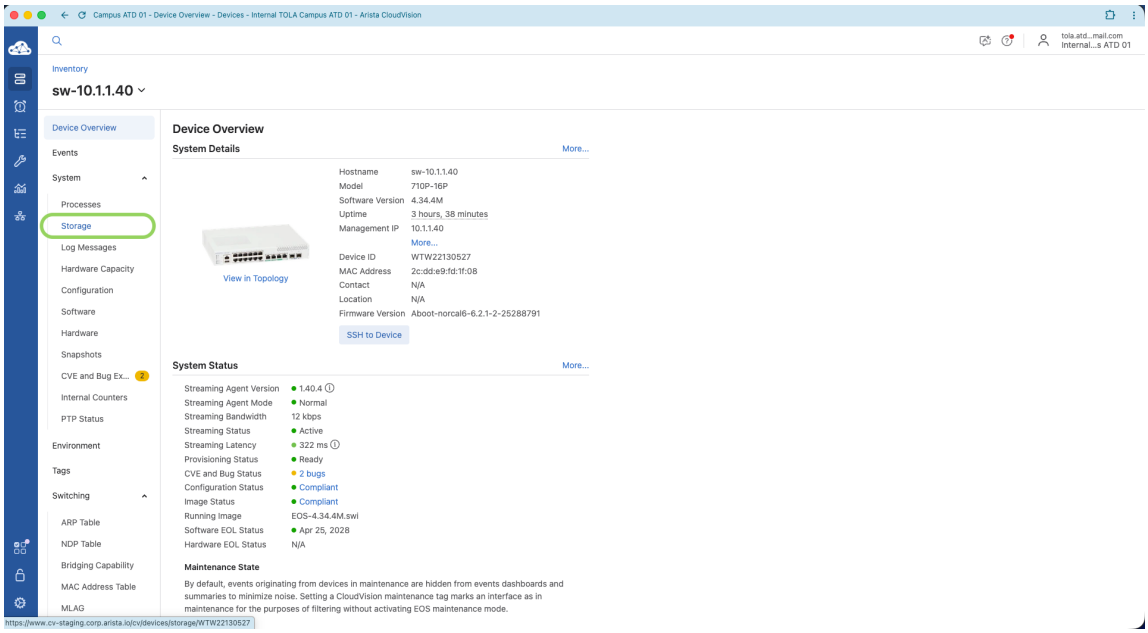
1. In CVP, Select Devices from the left hand menu



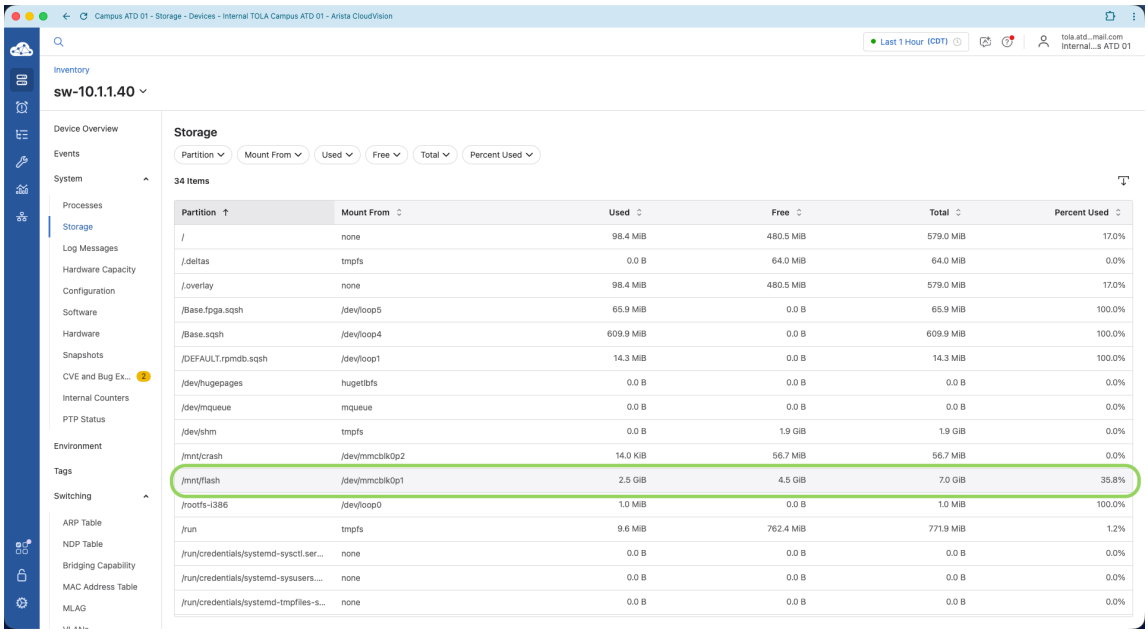
2. Verify the version of the switch then click on the switches `Hostname` in blue.



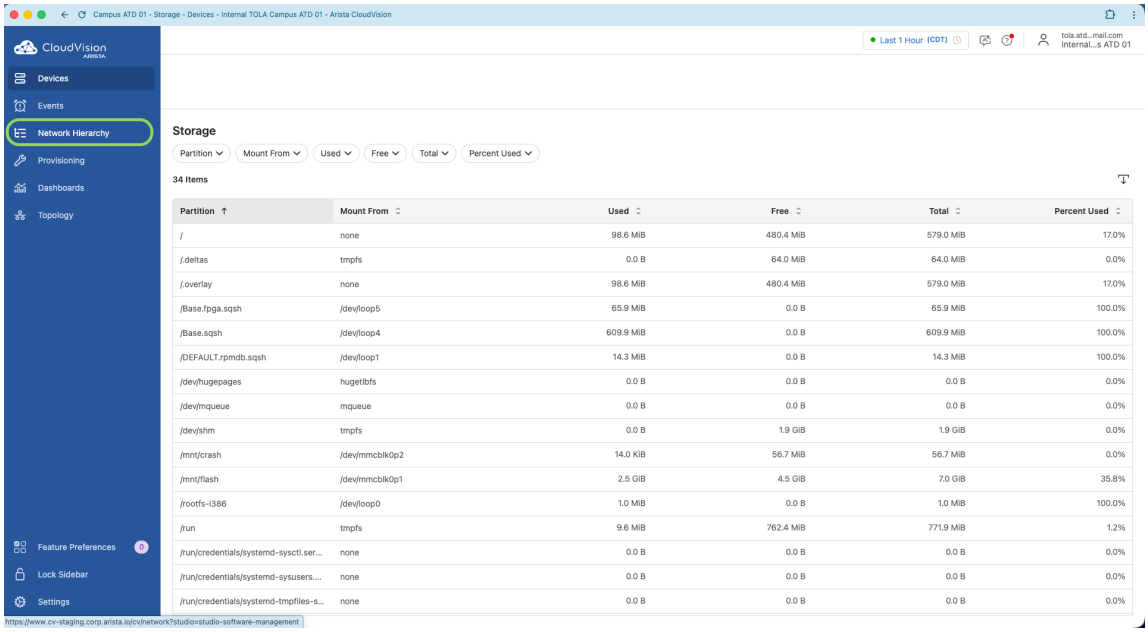
3. Next Select `Storage` from the `System` menu



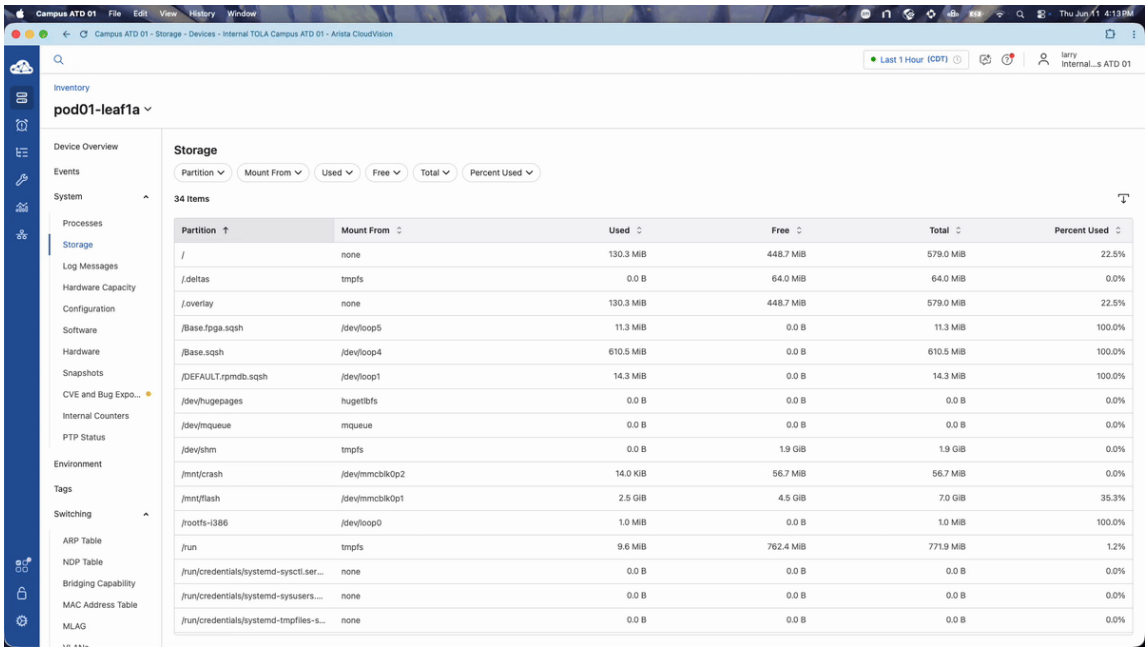
4. Notice how much free space is available on the switch



5. Next, navigate to Network Hierarchy



6. Select your IDF, and then the `Endpoints` window. Next, scroll down to find your RaspberryPi's IP address



7. Before we apply the new firmware, let's start a ping test which will run during the switch upgrade process. We will see that the ping traffic will continue to flow through the switch even while its software is being upgraded.

    a. Open a browser and type `http://<RaspiberryPi IP>:8501`

    b. Select the Ping App

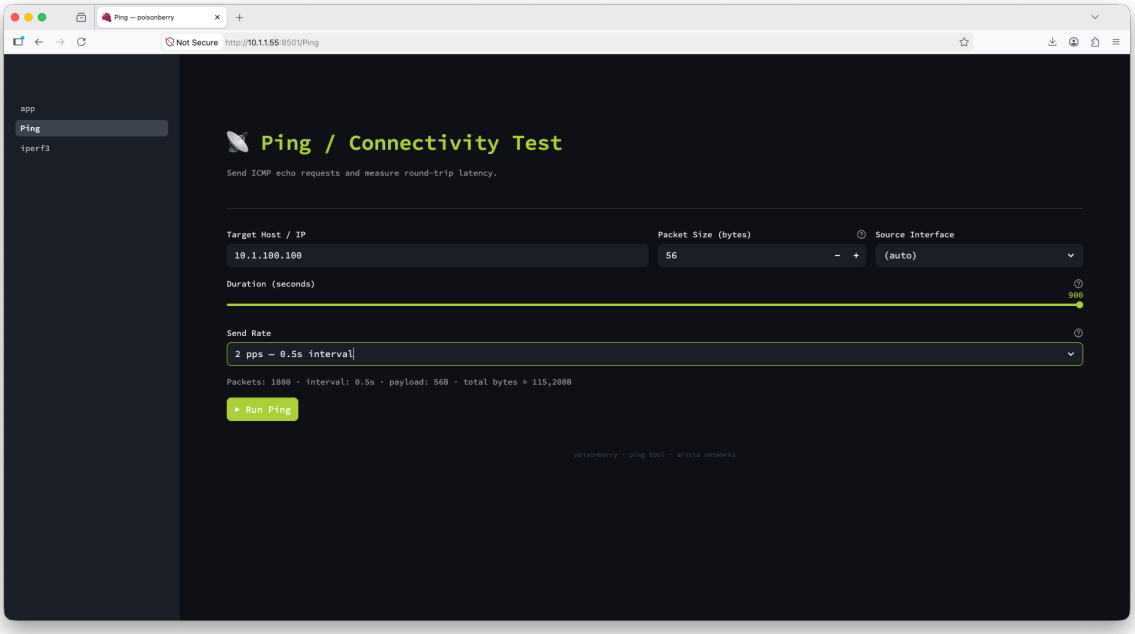
    c. Input the `Target Host / IP` for you POD's default Gateway: `10.1.##.1` or try Instructor Pi at `10.1.100.100`

    d. Change the duration to `900 seconds`

    e. Change the send rate to `2 pps - 0.5s interval`

    f. Select `Run Ping`

> **Ping Test**
>
> Let the `Ping/Connectivity Test` complete the 15 minute test to see the results. If you press stop during the test is will not display any results.



Alternative Ping Tests - Connect your personal device to the SSID created in the first wireless Lab - `ATD-###-PSK SSID`

MAC OS

Open Terminal and run the following, please replace ## with your pod number (1-12)

```
ping 10.1.##.1 -i .5
```

Windows

Open Command Prompt and run the following, please replace ## with your pod number (1-12)

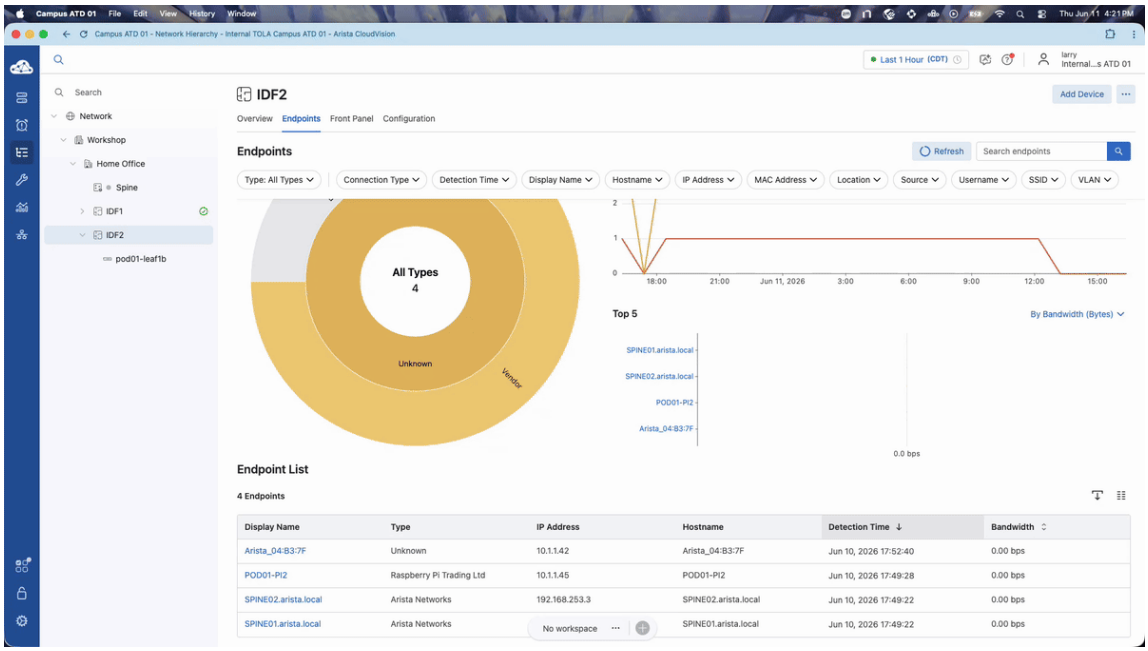
```
ping -t 10.1.##.1
```

Performing SSU

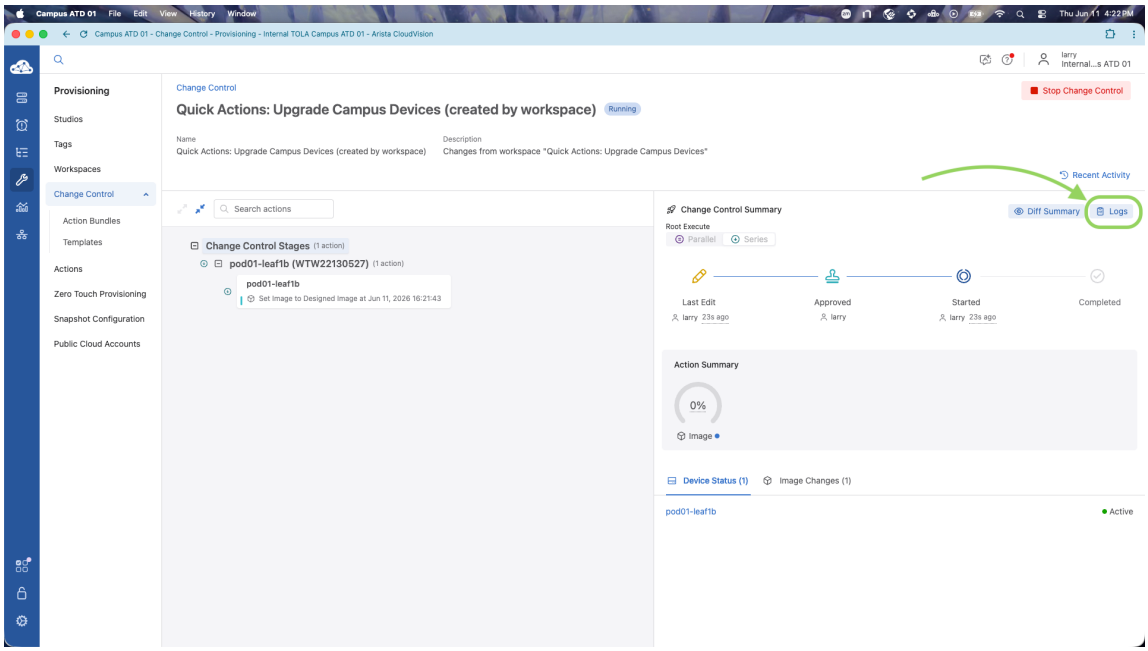
Now leave this window open for the following steps. We will see ping packets being sent and received every half second. You are now pinging the CORE switch IP address. The ping traffic must traverse the `leaf1a` or `leaf1b` switches and the `SPINES` to reach the gateway. We should be able

to observe how traffic is affected while the switch is upgrading during SSU.

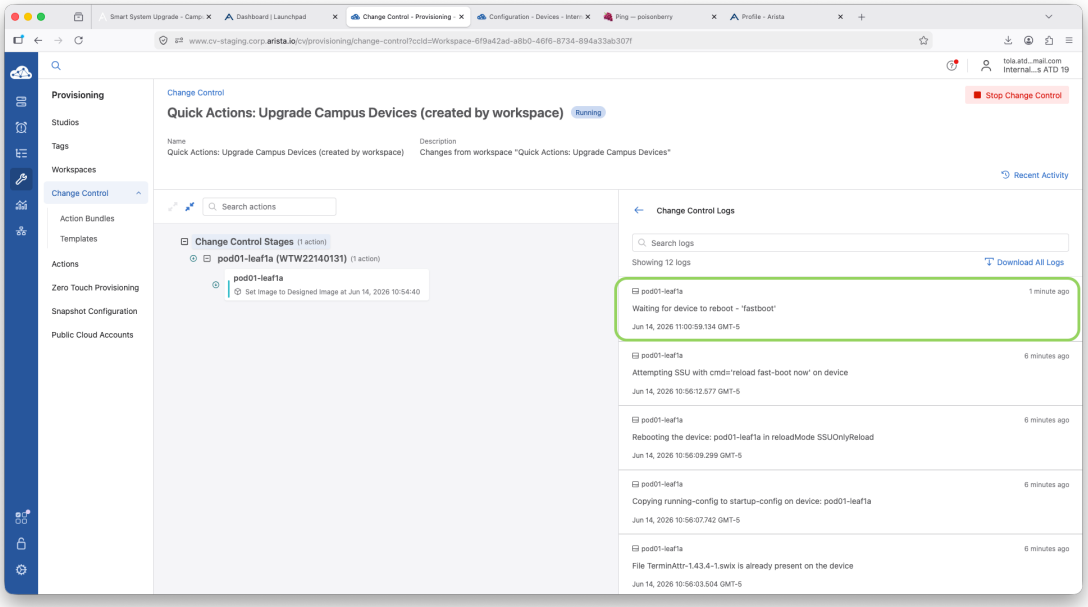
1. In a standard firmware upgrade process, you would issue a normal reload command. However, in this lab, we want to trigger an SSU upgrade. Issue the command below now.
2. Select your IDF then the 3 dots and select `Upgrade`
3. Select the Image 4.34.5M.swi and available streaming agent then continue
4. Select SSU then the drop menu. We are forcing SSU with no fall back, which will fail with any warnings. Select Apply
5. Select Continue
6. Confirm the Software Image and switch being upgraded
7. Select Upgrade



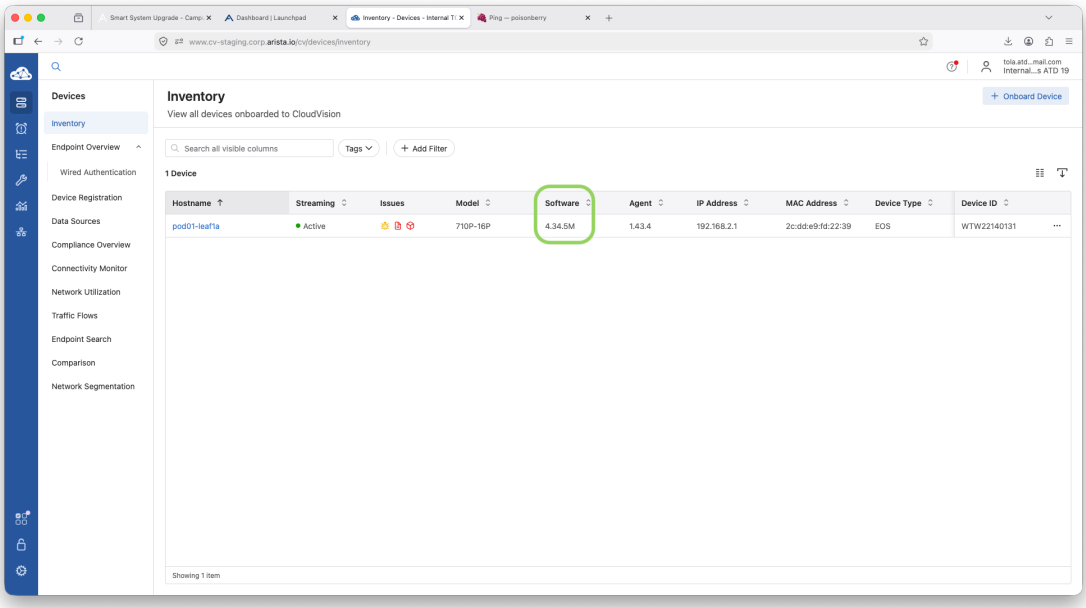
8. Select `View Change Control` as the SSU process proceeds. You can watch the progress from the logs. The switch will reboot shortly, and it will take some time before it begins streaming with CVP.



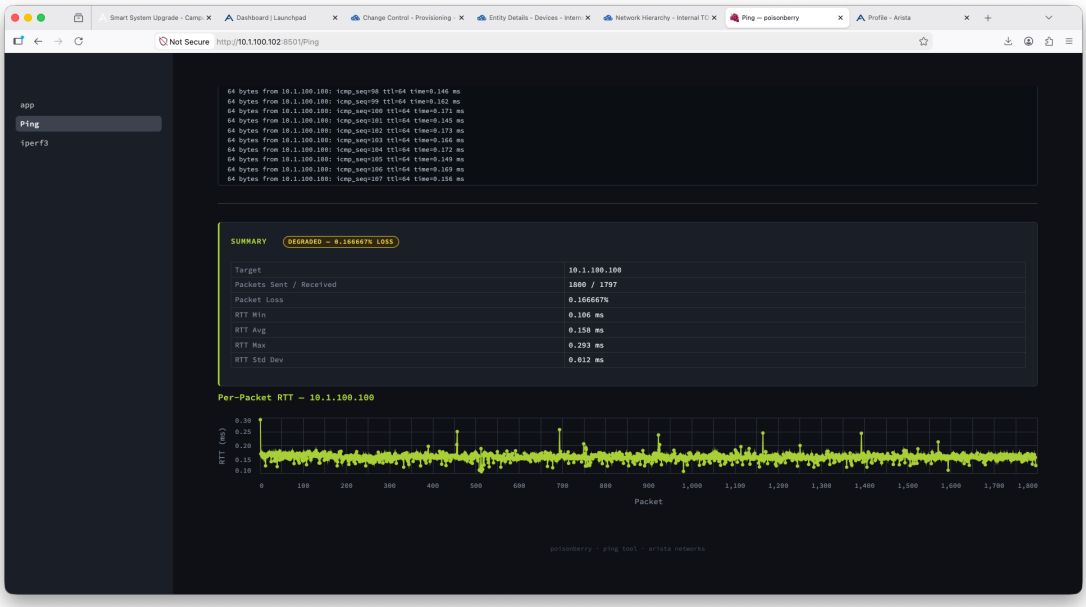
9. When you see the following message in the logs, the switch is rebooting.



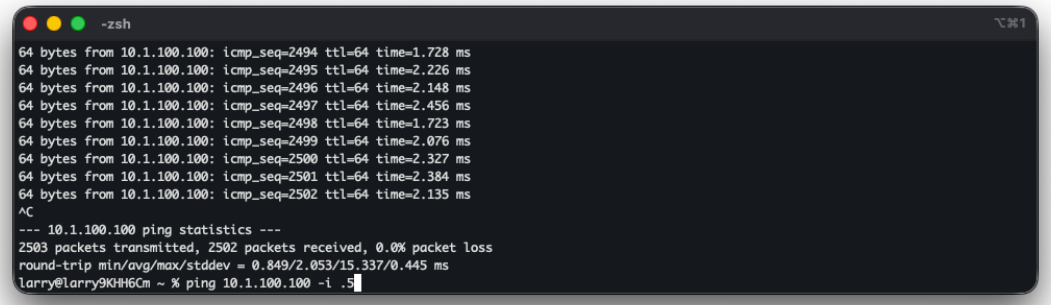
10. You can now verify that switch has been updated by going selecting the `Devices` menu . You should see the switch has upgraded to `EOS-4.34.5M`



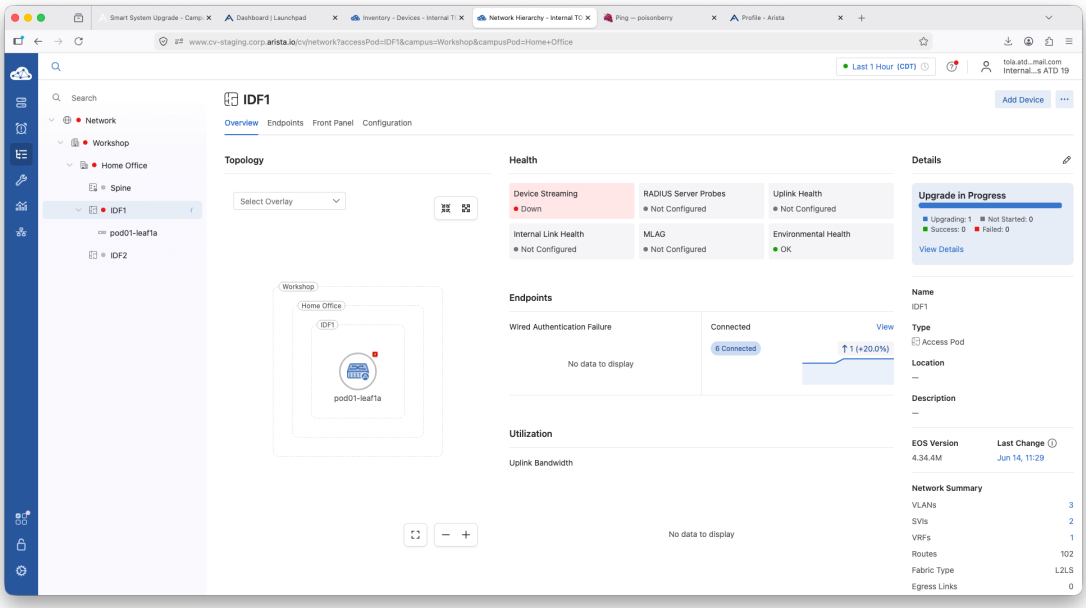
11. Take another look at `Poisonberry` that was running the consistent pings. You should see pings continue to flow without issue during the upgrade, and only a few pings were lost as the ASIC reconnects to the updated management plane.



12. The below example output was sending pings every 500 ms, and only `one` packet was lost. The cutover time to the new EOS image disrupted the dataplane for less than a second!! This is fast enough you may not even notice the disruption on an active zoom call!



13. Browse back over to Network Hierarchy and you'll notice your red dots in your hierarchy. You can see that `Device Streaming` had an issue. You can select any item in red to acknowledge the alert and return the site back to green/gray.



Lab Conclusion

We just observed how Arista SSU allows network connected devices to continue with little downtime to operate on the network even while an EOS firmware update occurs on the connected switch.

CONGRATS! You have completed the Wireless labs!

2026-06-14